12th HACKERS KA JUGAAD

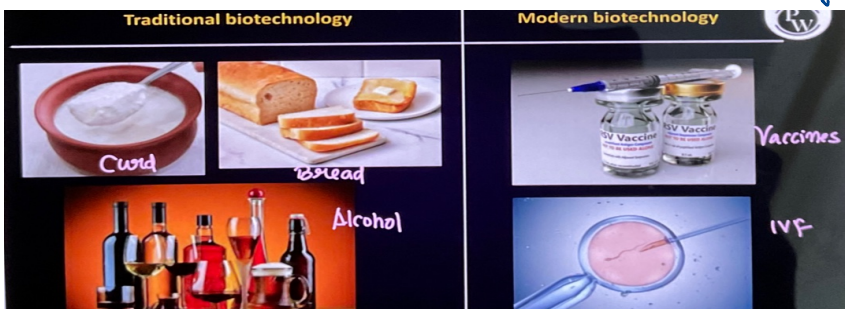
Biotechnology : Principles and Processes

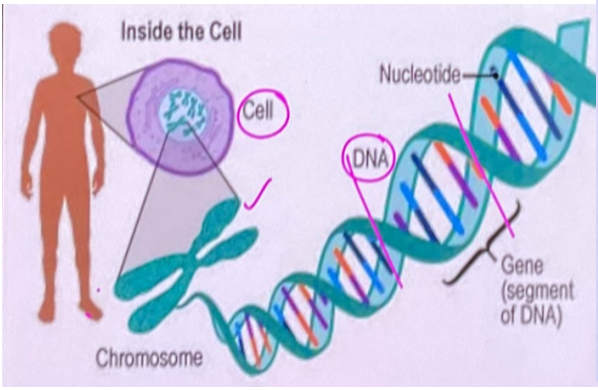
Biotechnology- techniques of using living organisms for products and processes, useful for humans

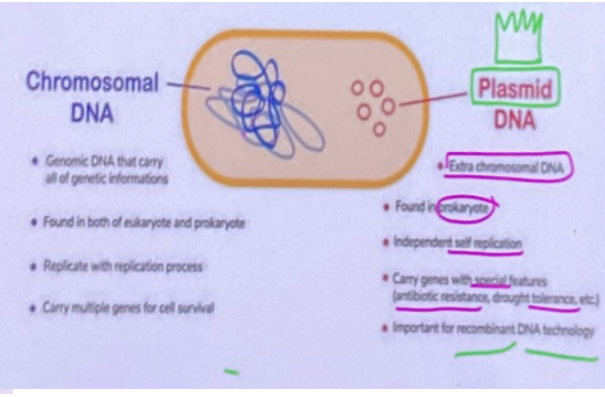
According to european Federation of biotechnology has given a definition of biotechnology, the integration of natural science and organisms, cells, parts there off and Molecular analogs for products and services .

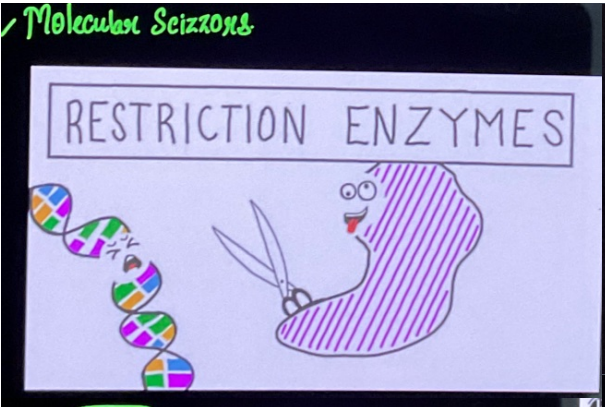
old and Modern Biotechnology

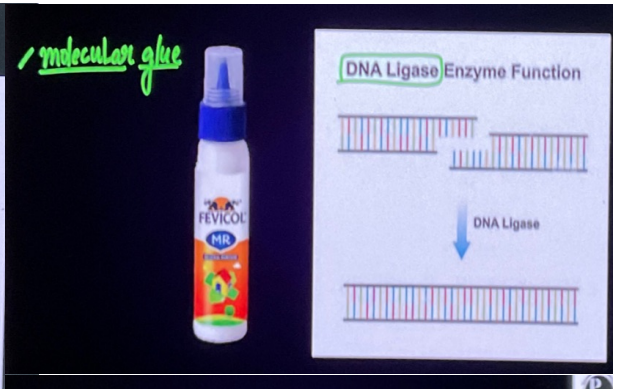
conventional /traditional/ old biotechnology	Modern biotechnology
– whole or complete organism is used	– use of genetically modified organism
– eg: curd, dosa, bread, beer, wine, etc.	– eg: vaccines, IVF, drugs like insulin, etc.

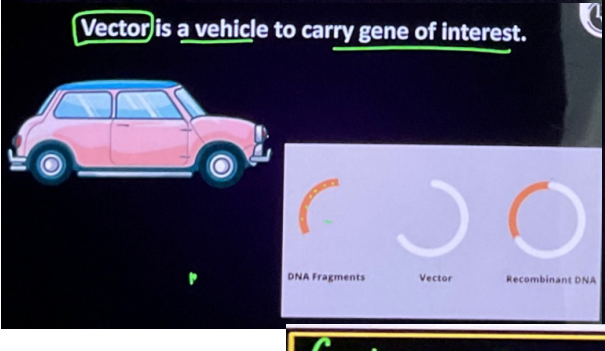


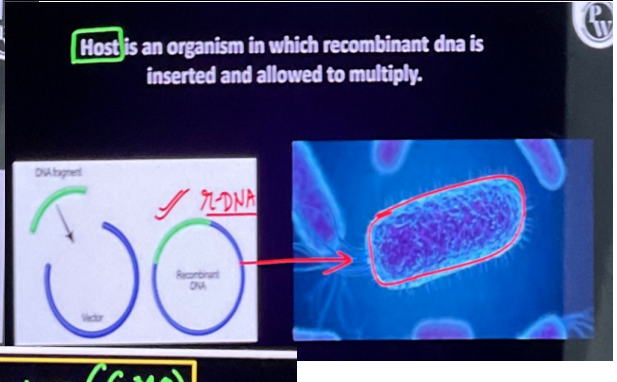

Inside the Cell
Cell
Nucleotide
DNA
Gene (segment of DNA)
Chromosome

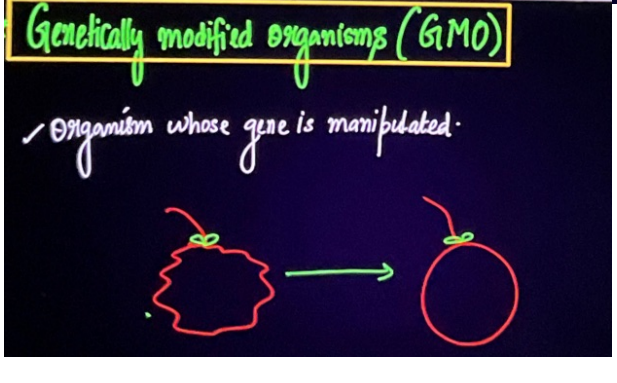

Chromosomal DNA
Plasmid DNA
Genomic DNA that carry all of genetic informations
Found in both of eukaryote and prokaryote
Replicate with replication process
Carry multiple genes for cell survival
Extra chromosomal DNA
Found in prokaryote
Independent self replication
Carry genes with special features (antibiotic resistance, drought tolerance, etc.)
Important for recombinant DNA technology


Molecular Seizzors
RESTRICTION ENZYMES


molecular glue
FEVICOL
MR
DNA Ligase Enzyme Function
DNA Ligase


Vector is a vehicle to carry gene of interest.
DNA Fragments
Vector
Recombinant DNA

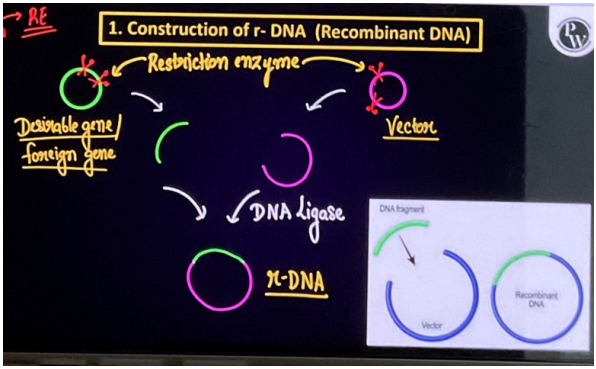

Host is an organism in which recombinant dna is inserted and allowed to multiply.
DNA fragment
rDNA
Vector
Recombinant DNA

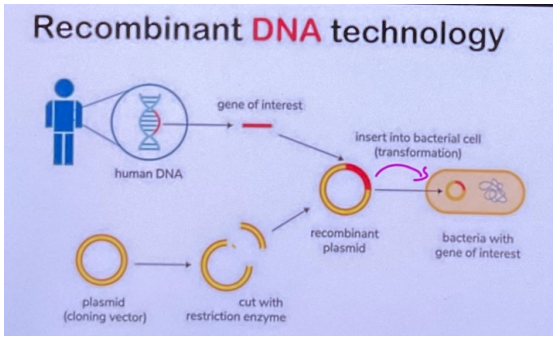

Genetically modified organisms (GMO)
Organism whose gene is manipulated.

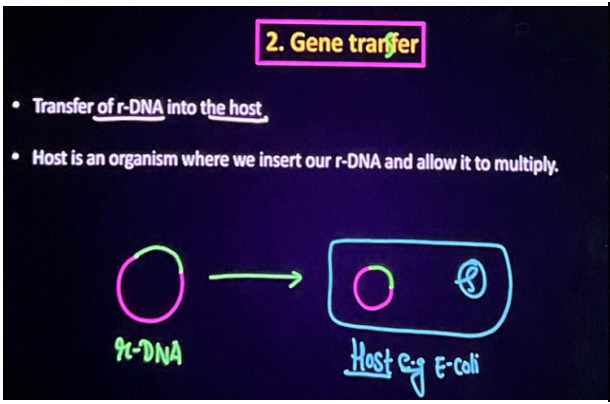
Principles of Biotechnology

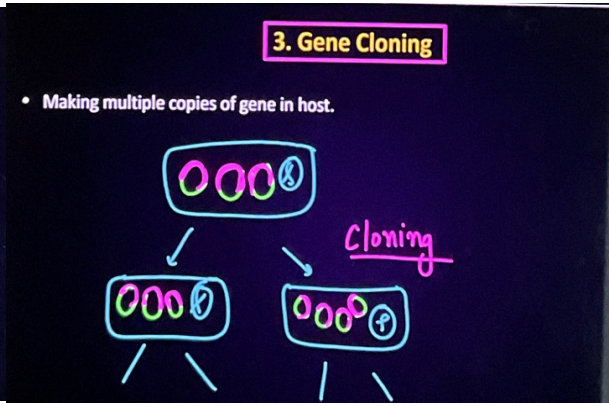
Genetic engineering
- Construction of RDNA
- Gene transfer
- Gene cloning

Biochemical engineering
- Providing a sterile environment that is contamination free to only enable the growth of desirable gene



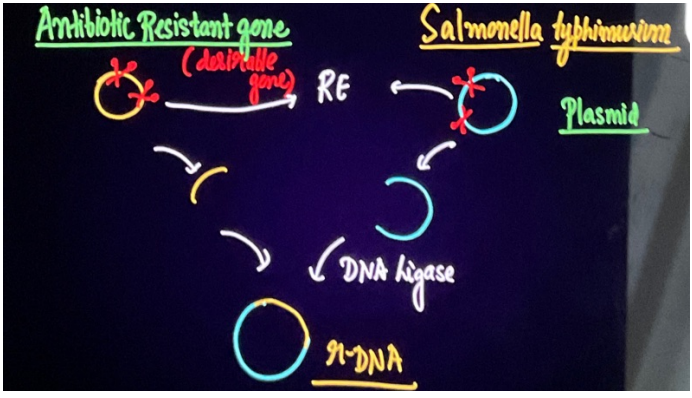






History of First Recombinant DNA

-First recombinant DNA was constructed in the year 1972
-By Stanley Cohen, and Herbert Boyer
-They did this by linking and antibiotic resistant gene with native plasmid of salmonella typhimurium



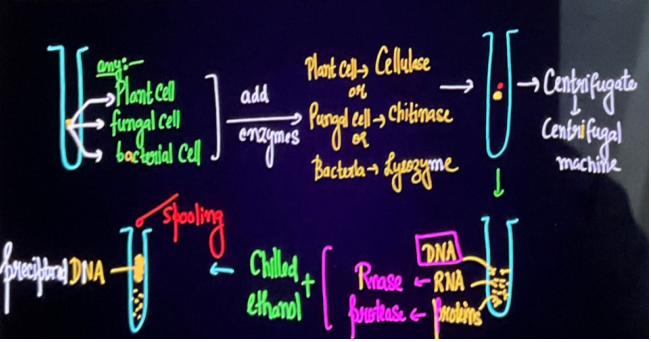
Three Basic steps in GMOs.

1. Identification of DNA with desirable genes
2. Introduction of identified DNA into the host
3. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny

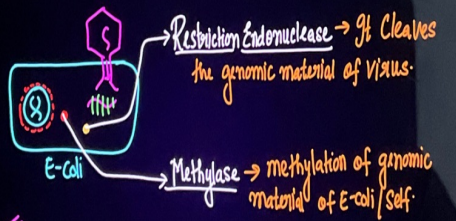
Steps involved in Biotechnology

1. identification and isolation of DNA
2. Fragmentation of DNA using restriction enzyme
3. Separation of the fragments using gel electrophoresis and isolating the desired gene
4. Multiple copies of desired gene are produced using PCR
5. Link the desired gene with vector
5. Transfer into host
6. Culture host in large amount
7. Downstream processing, product, isolation and marketing

1. Isolation of DNA (Genetic material)



2. Fragmentation of DNA using Restriction Enzyme

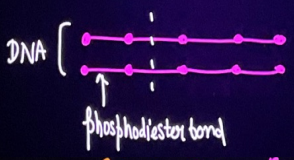


Note: In 1963 - 2 Enzymes were discovered within E-coli i.e. Restriction Endonuclease that cleaves viral genomic material and Methylase that adds methyl group to its own DNA (E.coli) to protect it.

RESTRICTION ENZYMES (RE)

Denoted as 'RE'

- These enzymes belong to the category of enzymes called Nucleases.
- Nuclease breaks phosphodiester bond.
- RE are obtained from Prokaryotes, that till date, we have more than 900 RE obtained from approx 230 Strains of Bacteria.



RESTRICTION ENZYMES (1963)

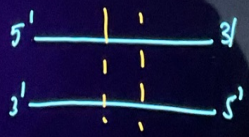
EXONUCLEASES

- It Removes the Nucleotide from the ends of the strand.



ENDONUCLEASES

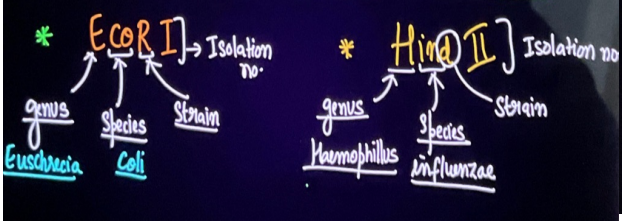
- It cuts the strand from the centre.



Naming of RE

Examples of RE:- EcoRI, BamHI, ClaI, Hind II, Hind III etc.

- 1st Restriction enzyme was Hind-II which was isolated and characterised 5 yrs later.



Working of RE

Recognition site

- Palindromic Sequence → RE cuts DNA Strand at a particular location

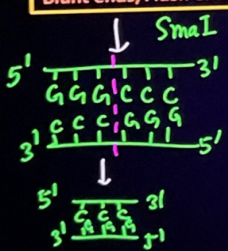
Eg MOM, MALYALAM

eg:- EcoRI → 6 base pair sequence "GAATTC"

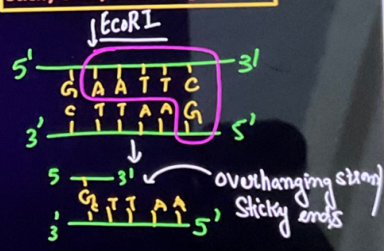
Hind II → 6 base pair sequence

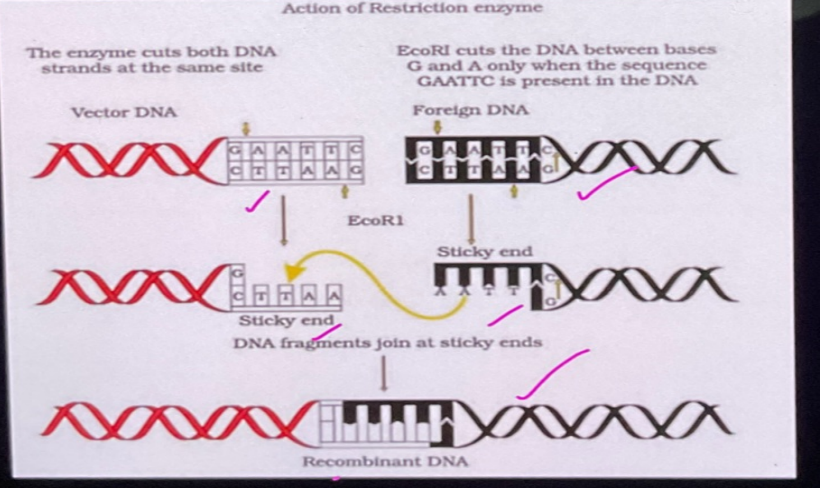
Endonuclease makes two types of cut ends

Blunt ends/Flush ends



Sticky ends/Over hanging ends





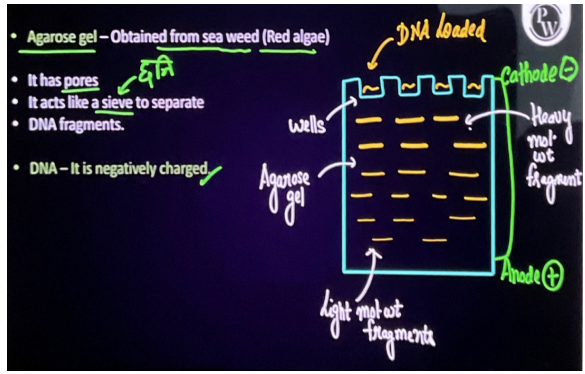
Function of RE -

- inspecting length of DNA
- Binds to specific recognition sequence
- Cut the two strands of ds DNA at specific points.

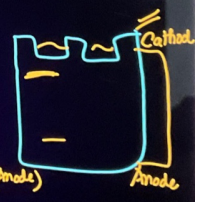
3. Separation of DNA with Gel Electrophoresis for isolation of Desired gene

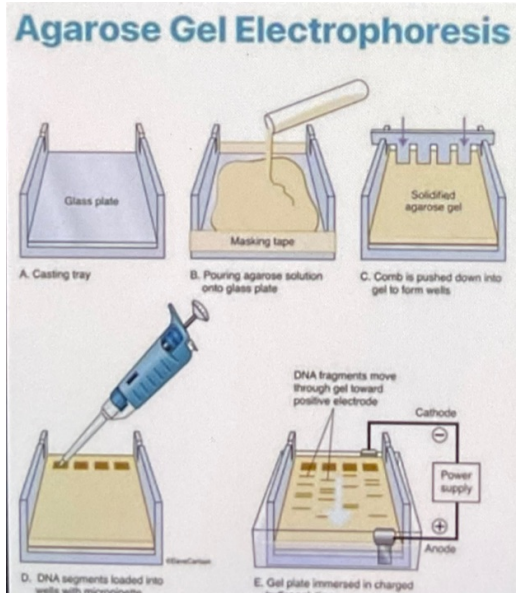
- Gel Electrophoresis
- Principle - on the basis of weight/Size – Separation
- Raw material
  - Agarose gel
  - Ethidium Bromide
  - Separation comb
  - U.V light
  - Electrodes (Cathode & Anode)

- Agarose gel – Obtained from sea weed (Red algae)
- It has pores
- It acts like a sieve to separate DNA fragments.
- DNA – It is negatively charged



- Technique – Pouring Agarose gel in glass box
- Using Separation comb plates well created
- DNA fragments loaded in wells (cathode side)
- Electrodes connected
- When separation starts
- Heavy DNA fragments – move slow
  Light DNA fragments – Move fast (towards Anode)
- Staining DNA fragments with Ethidium Bromide
- Visualisation under U.V light – DNA fragments appear orange
- Gene of Interest is extracted - Elution



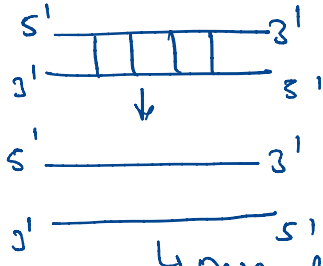


4. Multiplication/Amplication of Gene of Interest by PCR

- PCR (Polymerase chain Reaction)
- Teq used to multiply gene of Interest.
- 3 Steps - i) Denaturation ✓
  ii) Annealing ✓
  iii) Extension ✓

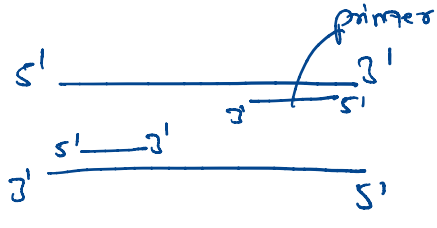
Raw material -
- PCR machine ✓
- Oligonucleotide primer ✓
- Enzyme – Taq. polymerase ✓

1.) Denaturation
≈ 92°C ~ 95°C

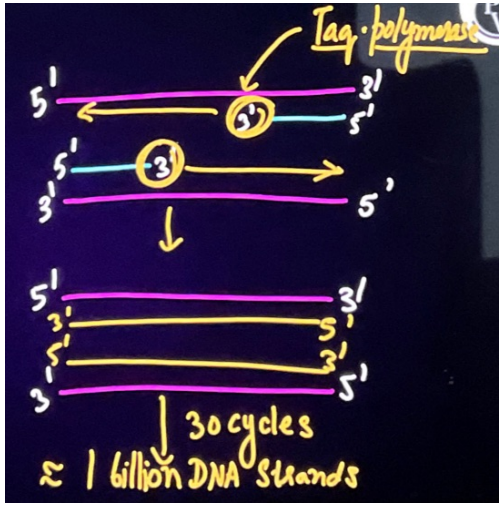


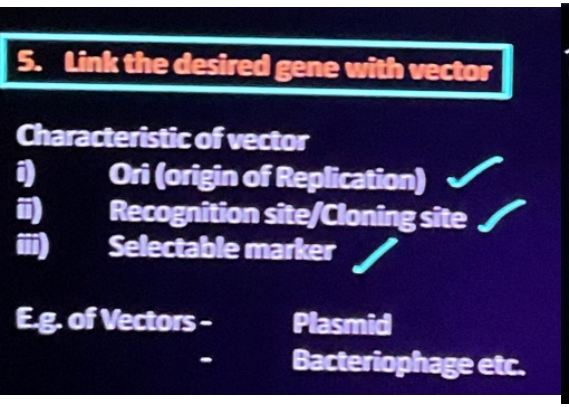
↳ DNA fragments separated.

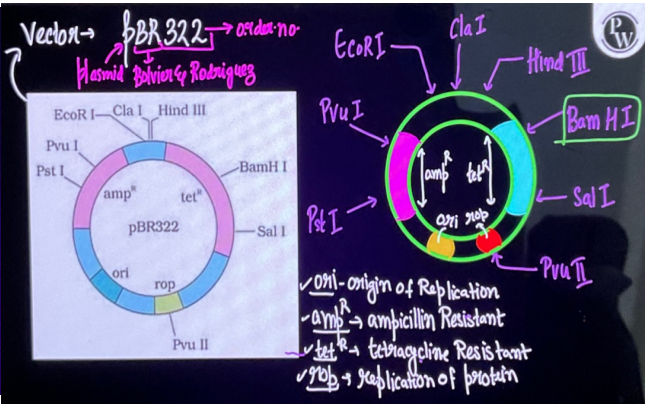
2.) Annealing
≈ 55°C
oligonucleotide primer
↓
attaches at free 3'-OH end.



3.) Extension
≈ 72°C
enzyme → taq. polymerase
↓
thermo aquaticus
– thermostable







Origin of replication

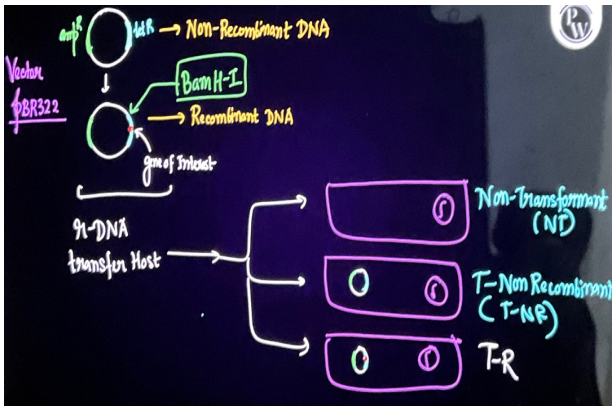
- ORI is a site from where replication start
- ORI is important in a vector as gene of interest when links with vector needs to be replicated
- Controls copy number

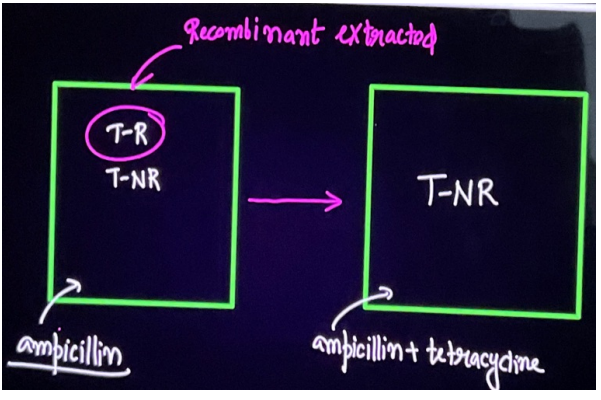
Cloning site or recognition site

- This is a site which is recognised by RE for cutting
- Restriction site has palindromic sequence
- If a vector has multiple recognition site for different enzyme-good
- If a RE has multiple recognition site on a single vector-bad

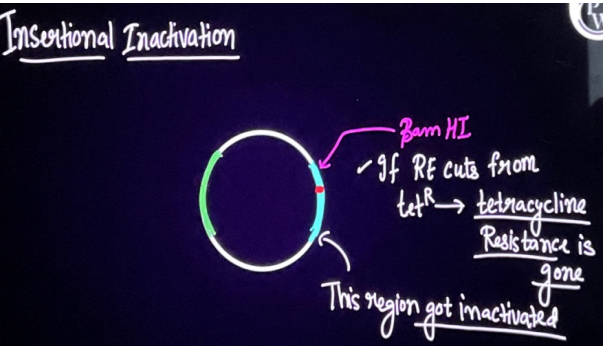
Selectable marker

- It helps in distinguishing between recombinant DNA and non recombinant DNA
- Also helps in distinguishing between transformant and non transformant

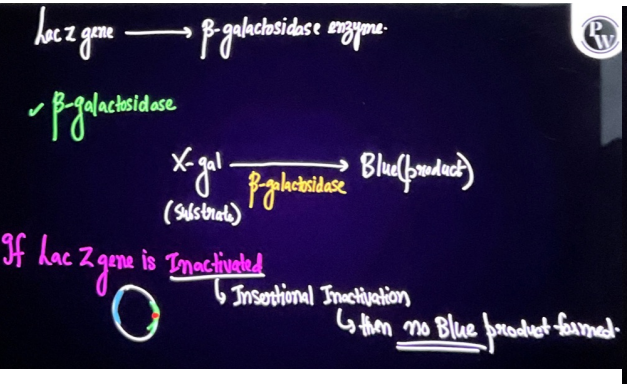


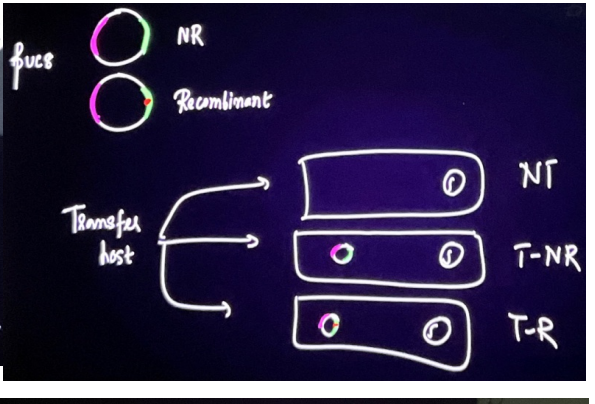


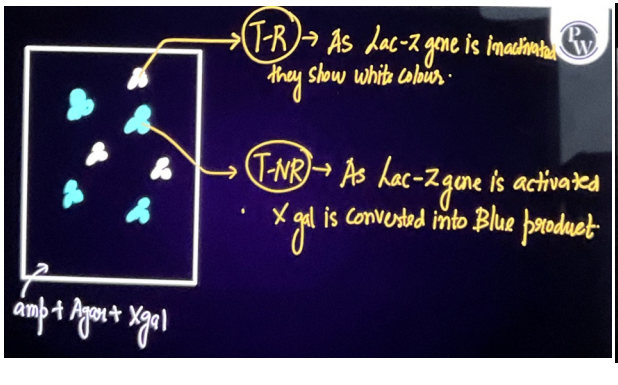
Insertional Inactivation

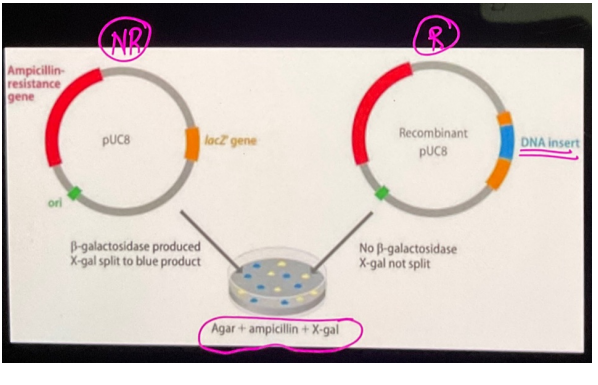








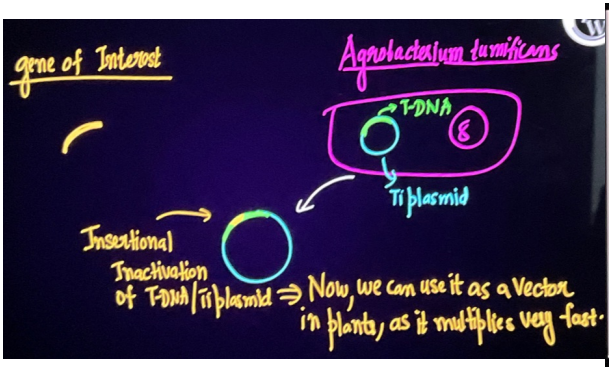


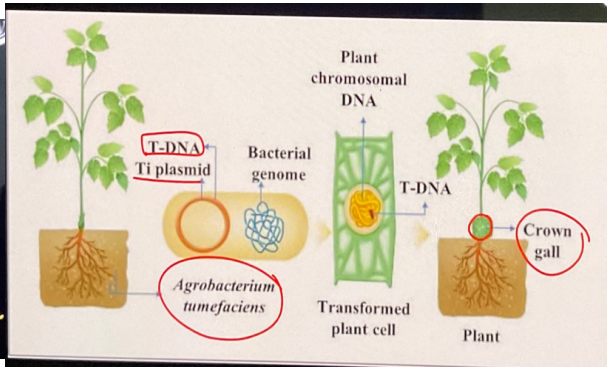


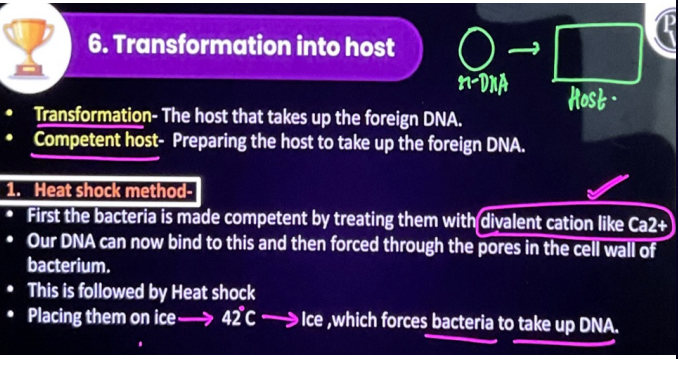
Vectors for cloning genes in plants and animals

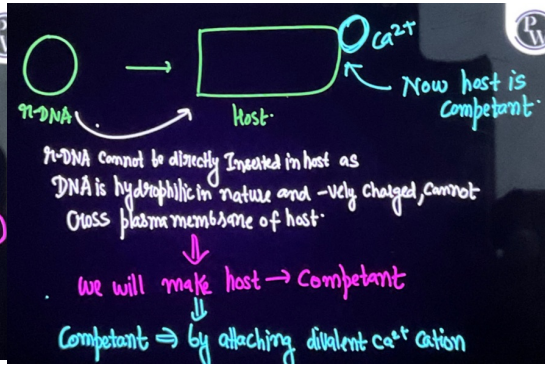
1.Agrobacterium Tumifaciens is a pathogen of several dicot. Plants able to deliver a piece of DNA known as TDNA to transform normal plant cell into a tumour cell and direct the tumour cell to produce the chemicals required by the pathogen TI plasmid is disarmed and t dna is used to deliver desirable genes

2.Similarly, retrovirus have also been disarmed and now used to deliver genes to animal cells. So once a gene or a dna fragment has been ligated into a suitable vector it is trasferred into bacteria plant or animal host where it multiplies









2. Gene gun/Biolistics

- Usually performed for transformation of plant cell
- DNA is coated on gold/Tungsten
- And is bombarded with high velocity to the plant cell.

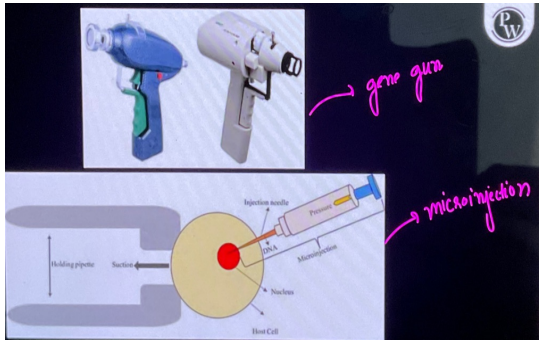
gold/Tungsten

host

3. Microinjection

- Usually performed for transformation of animal cell.
- Using microneedle/microinjection DNA or r-DNA is directly injected into the nucleus of host cell.

4. Disarmed vectors like retrovirus and *Agrobacterium tumifaciens* can be used directly to transform host.

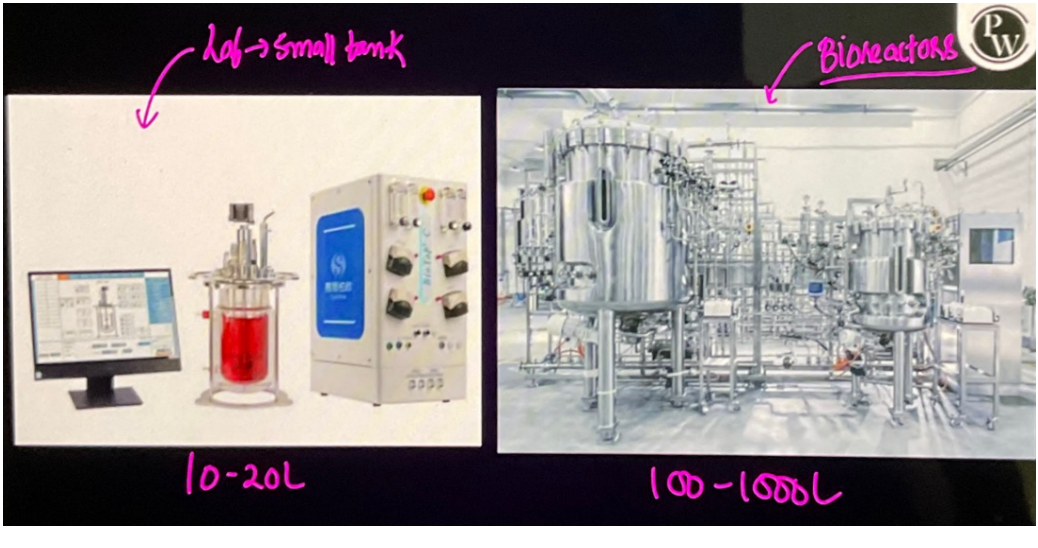


Culture host in large amount

- **Bioreactors:** To obtain the product on a large scale, big tanks called bioreactors are used to culture the host in large amount upto (100-1000 L)
- Prior to this small tanks, pilot plants upto 10-20 L capacity is runned initially to understand all conditions required for the culture.

- **2 types of bioreactors:**

1. Simple stirred bioreactor: Base curved
2. Sparged stirred bioreactor: Base flat, sparged tube for more oxygen availability.



properties of bioreactor

1.) Large amount of host can be cultured.

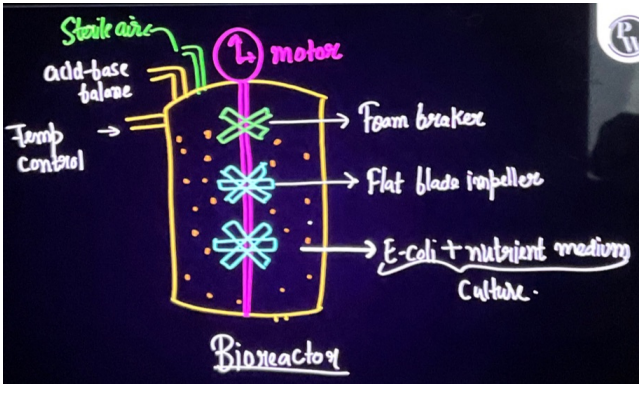
2.) Acid base balance control system.

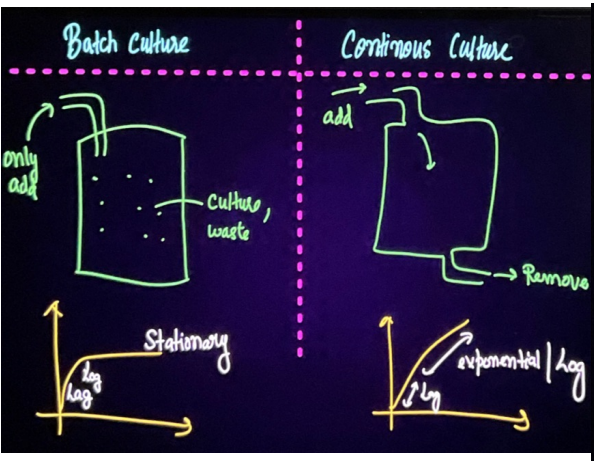
3.) Temperature control system

4.) foam breaker

5.) Sterile conditions maintained to avoid contamination

6.) Air availability via agitator system.





Batch culture	Continous culture
• Here once the host and nutrient media is added in the bioreactor it is closed and no further addition or removal waste can be done.	• Here we can add fresh nutrient medium in between process can be added and metabolic waste can be removed.
• Here growth shows lag exponential and stationary phase as further addition cannot be done.	• Here after lag phase, exponential phase increases as fresh media can be added and metabolic waste can be removed.

Downstream Processing

- Once culture is grown in bioreactor (e.g- E.coli) then the product (e.g- insulin) is isolated, separated and purified is called downstream processing.
- Separation and purification of the desired product/recombinant protein from heterologous host (non native host).
- Product has to be formulated with suitable preservatives.
- Strict quality control testing is done for each product
- The downstream processing and quality control testing vary from product to product.

THANK You !!